



Programming Languages



At the end of this lesson, you should be able to:

- Distinguish between the low-level machine-language and high-level programming language
- Describe the two ways of translating high-level language program code to machine-language instructions
- State the main features of high-level programming languages: C, C++, Java, Python

Programming is the process of

- implementing a representation of the solution for the computer to execute
- taking an algorithm and encoding it using certain programming language

Programming language

- a medium through which programmer may give instructions to a computer

A programming language

- must support certain control constructs and data types needed to implement algorithms



Examples

- Sequential processing
- Decision-making selection
- Iteration for repetitive execution
- Numbers such as integer and real (floating point)
- Characters and Strings

Types of Programming Language

The computer has its own language.

- **machine-language** that is specific to the type of computer processor
- consists of a set of machine instructions and data objects that are encoded in '0' and '1' binary format
- error-prone for human to code a program directly using the computer's machine codes



Types of Programming Language (Cont'd)

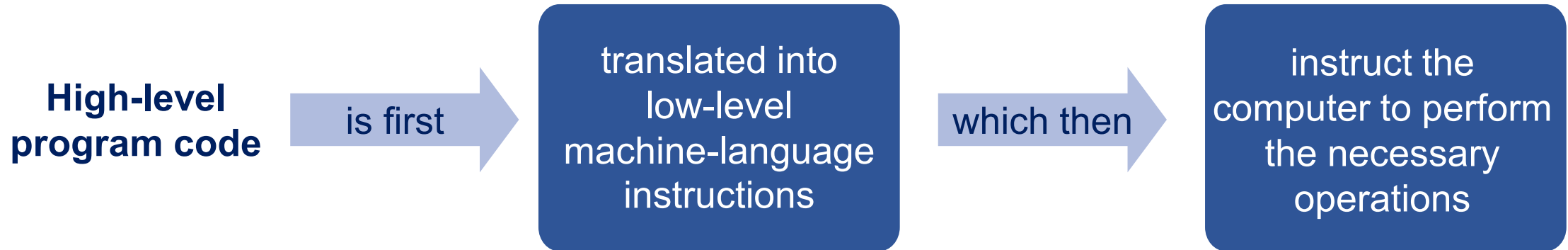
High-level programming languages that are more user-friendly are hence invented.

- provide abstraction from the internal operating detail of the computer
- enable the programmers to focus on solving the problem
- make the process of developing the algorithm simpler

Most programs used in modern computers

- coded in high-level programming languages, such as C, Java, Python, etc.



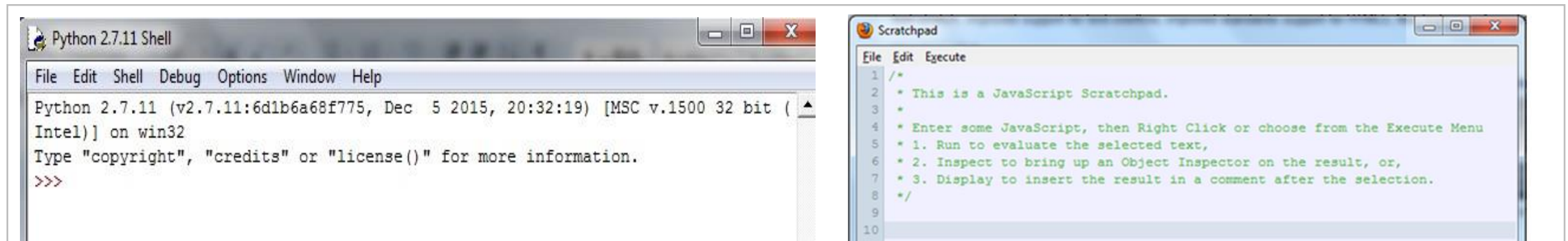


Two approaches for program translation:

- interpretation
- compilation

Interpretation approach

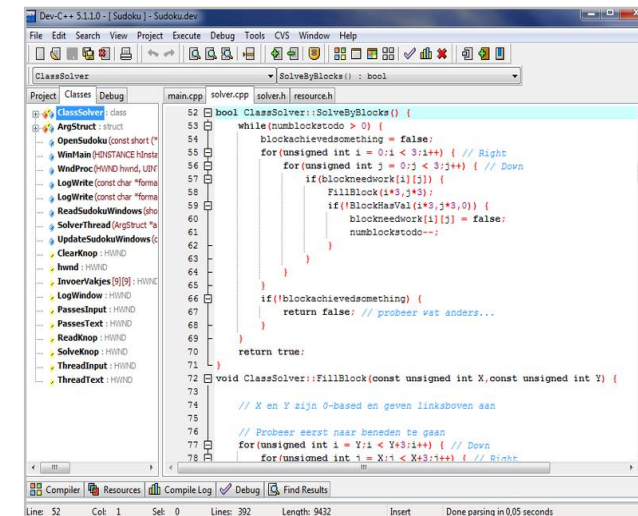
- uses a program known as **interpreter**
- reads one high-level code statement at a time
 - immediately translates and executes the statement before processing the next one
- examples: Python, R, and JavaScript



Program Translation: Compilation Approach

Compilation approach

- uses a program called **compiler**
- reads and translates the entire high-level language program (source) code into its equivalent machine-language instructions in an executable file
- the resulting machine-language instructions can then be executed directly on the computer when the program is launched
- examples: C and C++



It is also possible to use the **combination of both** translation techniques

- example: Java

Interpreter vs. Compiler

Interpreter

- very portable across different computing platforms
- produces results almost immediately
- easy to debug
- program executes more slowly
- useful for implementing dynamic, interactive features, such as those used on web pages

Compiler

- program runs very fast AFTER compilation
- smaller in code size after compilation
- must compile the entire program before execution
- needs to be re-compiled if to be used on different computing platforms
- used in large and sophisticated software applications when speed is of the utmost importance

Programming Language: C

C language

- originally developed to help implement the Unix operating system
- allow for direct access of, and manipulation of, the underlying computer's hardware

It is a compiled language

- the program code (i.e. source code) needs to be compiled first, before it can be executed

Most suitable for applications

- direct access to the computer's hardware
- fast real-time response



Examples of applications

- embedded systems development
- operating systems implementations
- device drivers that control the hardware inside the computer

C++ language

- an extension to the C language
- with additional support for Object-Oriented Programming (OOP)

In OOP

- programs are designed based on objects that contain attributes and behaviors
- programming is then focused on how the objects interact with one another
 - without the need to know the internal code detail of the object

C++ is a compiled language

- used for applications that need high performance and are based on object oriented software design principles



Examples of Applications

- Video games
- Graphic User Interface (GUI) based applications
- Latest operating systems design and implementation

Python language

- a relatively easy-to-learn programming language
- targeted for general purpose programming for applications that do not need direct access to the computer's hardware

It is an interpreted language

- program code is hence portable across different computing platforms

Python interpreter converts and executes the Python program statement one-by-one to the corresponding machine instructions

- provides instant feedback to the user at runtime
- very helpful for learning to program



Examples of Applications

- Data science
- Web developments

Java language

- highly portable, general purpose language with Object-Oriented (OO) methodological support

Designed to run as an interpreted language on a Java Virtual Machine (JVM)

- portable across different types of machines and devices
- no direct access to the computer's hardware

Combination of compiler and interpreter, as well as Just-in-time (JIT) compiler

- the source code (with file extension .java) is first compiled to Java bytecode (with file extension .class)
- Java bytecode instruction is then interpreted by the JVM interpreter during execution
- JVM may call the JIT compiler to compile some of the bytecode instructions at runtime before they are executed, achieving better performance



Examples of Applications

- Mobile Apps for Android phones
- Game development

Programming Language: Examples of Source Code

Source Code of a C Program

```
#include <stdio.h>
int main()
{
    printf("Hello, World!");
    return 0; //success
}
```



Source Code of a Java Program

```
public class HelloWorld {
    public static void main( String[] args ) {
        System.out.println("Hello, World!");
        System.exit( 0 ); //success
    }
}
```



Source Code of a C++ Program

```
#include <iostream>
int main()
{
    std::cout << "Hello, World!";
    return 0;
}
```



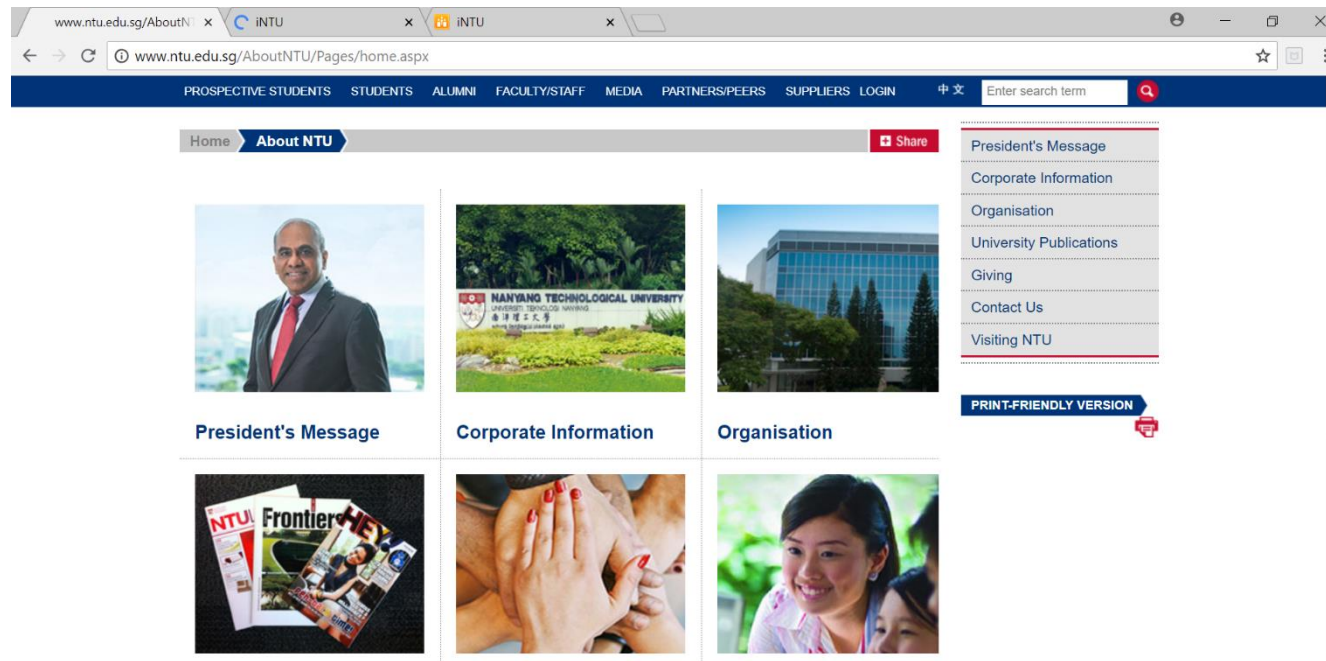
Source Code of a Python Program

```
print('Hello, World!')
```



Programming Language for Internet

Webpage: a text document transferred over the internet and be displayed in a **web browser**, such as Chrome, Safari, Firefox, and Internet Explorer (IE)

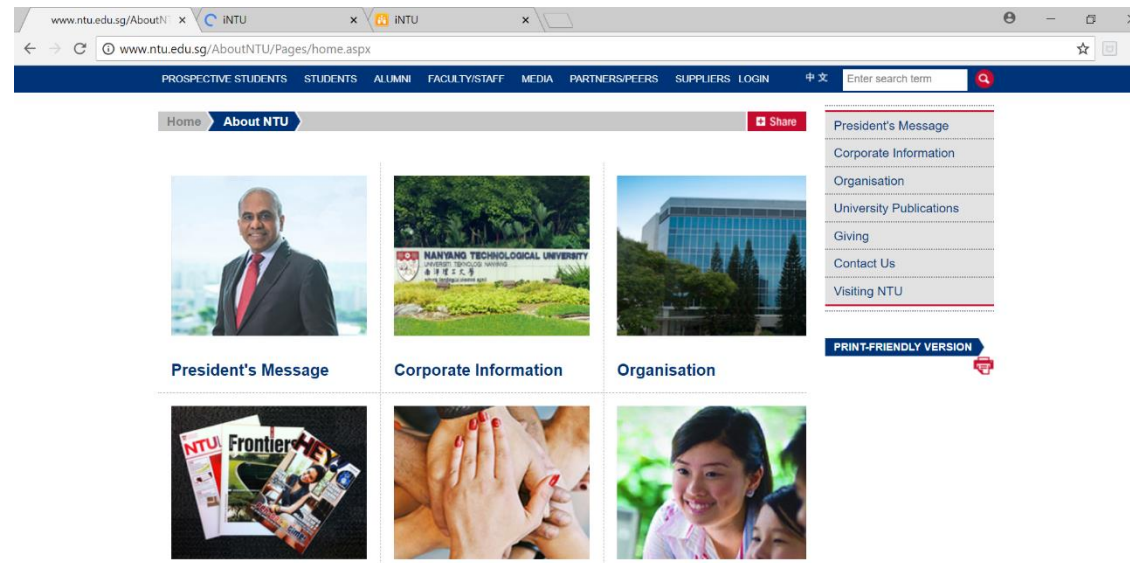


A **web browser** reads the webpage and composes it into visible or audible page(s).

Programming Language for Internet (Cont'd)

The content of a webpage is commonly described by using the language **HTML** (Hypertext Markup Language)

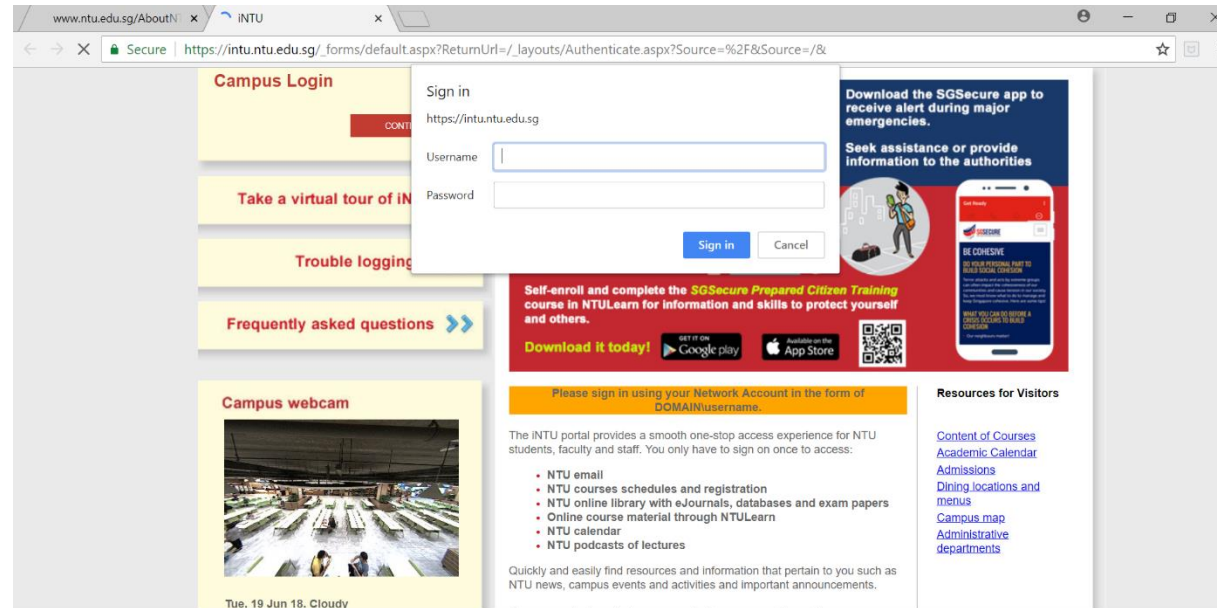
- a content-markup language uses symbols and phrases
- instructs the web browser to structure information for display or process



Programming Language for Internet (Cont'd)

A webpage can be made to appear dynamic and interactive by adding scripts in language, such as **JavaScript**.

Example: to prompt for user's name in the webpage



PHP (PHP: Hypertext Pre-processor) is another script language



- dynamically generates html webpage before it is sent to the browser
- Example: contents that need to be composed based on user's request



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CZ1003 INTRODUCTION TO COMPUTATIONAL THINKING* 3.0 AU

INDEX	TYPE	GROUP	DAY	TIME	VENUE	REMARK
10114	LEC/STUDIO	CS1	MON	1030-1130	LT2A	
	LEC/STUDIO	CS1	THU	1130-1230	LT2A	
	LAB	DSAI1	MON	1430-1530	HWLAB2	
	LAB	DSAI1	MON	1530-1630	HWLAB2	
10115	LEC/STUDIO	CS1	MON	1030-1130	LT2A	
	LEC/STUDIO	CS1	THU	1130-1230	LT2A	
	LAB	DSAI2	WED	1230-1330	HWLAB2	
	LAB	DSAI2	WED	1330-1430	HWLAB2	
10116	LEC/STUDIO	CS1	MON	1030-1130	LT2A	
	LEC/STUDIO	CS1	THU	1130-1230	LT2A	
	LAB	FS1	FRI	1430-1530	SWLAB1	
	LAB	FS1	FRI	1530-1630	SWLAB1	

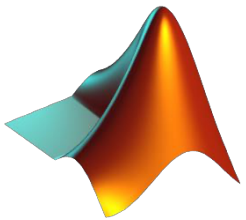
Programming Language: Others

R language



- an interpreted language
- used for statistical computing, such as data analysis
- provides extensive graphic plotting functions
 - useful for displaying data trends

MATLAB language

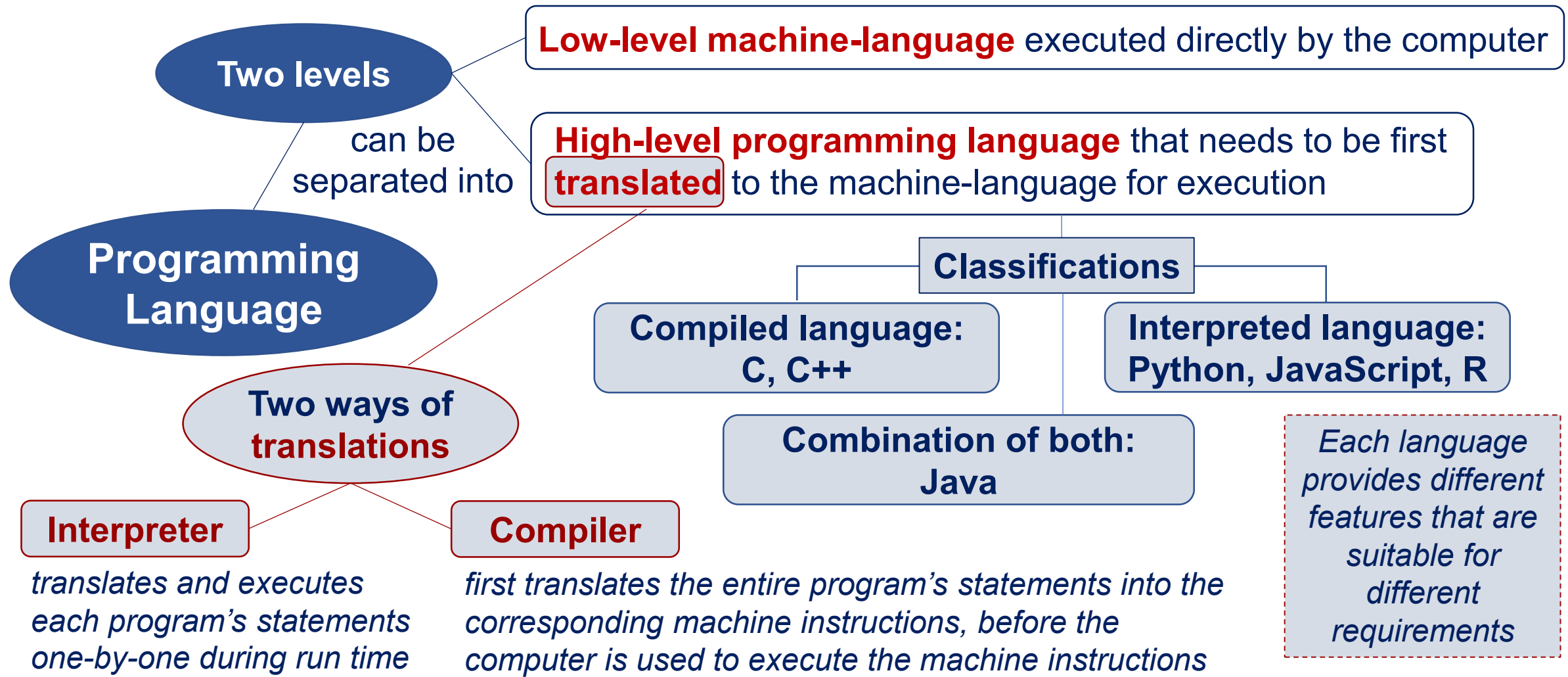


- commonly used for developing science and engineering computation applications
- an interpreted language used primarily for numerical computing
- supports many functions and libraries, such as matrix manipulation, plotting, and visualization of data

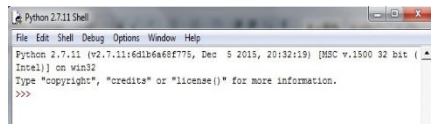
Examples of types of computation:

- audio, video, and image related signal processing
- control systems

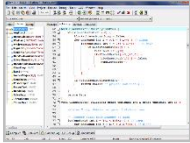
Summary



References for Images

No.	Slide No.	Image	Reference
1	3		Monitor screen [Online Image]. Retrieved June 20, 2018 from https://pixabay.com/en/monitor-screen-computer-speaker-2455524/ .
2	6, 13, 15		Python logo [Online Image]. Retrieved June 20, 2018 from https://www.python.org/static/community_logos/python-logo-master-v3-TM.png .
3	6, 14, 15		Java log [Online Image]. Retrieved June 20, 2018 from http://www.oracle.com/us/technologies/java/java-licensing-logo-guidelines-1908204.pdf .
4	8		By Annakoppad - Own work, CC BY-SA 4.0, retrieved June 20, 2018 from https://commons.wikimedia.org/w/index.php?curid=48721101 .
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References for Images

No.	Slide No.	Image	Reference
6	9		Devcpp5110 [Online Image]. Retrieved June 20, 2018 from https://pixabay.com/en/amd-cpu-processor-microprocessor-1310766/ .
7	10, 15		By Jeremy Kratz - https://github.com/isocpp/logos , Copyrighted free use, retrieved June 20, 2018 from https://commons.wikimedia.org/w/index.php?curid=62851110 .
8	17		By W3C, CC BY 3.0, retrieved June 20, 2018 from https://commons.wikimedia.org/w/index.php?curid=12736763 .
9	19		Php Logo [Online Image]. Retrieved June 20, 2018 from http://php.net/images/logos/new-php-logo.png .



Introduction to Computational Thinking: Course Introduction and Overview

What is Computational Thinking?

Problem Solving Process



- formulates the solution to a problem in the form of algorithms
- solves by using computer
- consists of several parts

A L G O R I T H M

- a sequence of step-by-step instructions
- tells the computer how to solve a problem
- used extensively in many real-world applications

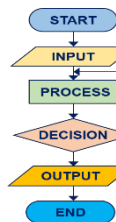
OUTLINE



Overview of
Programming
Language



Computer
Internal
operations



Basic Program
Structure and
Construct



Computational
Thinking
Process



Algorithm
Complexity



Computing
Trends

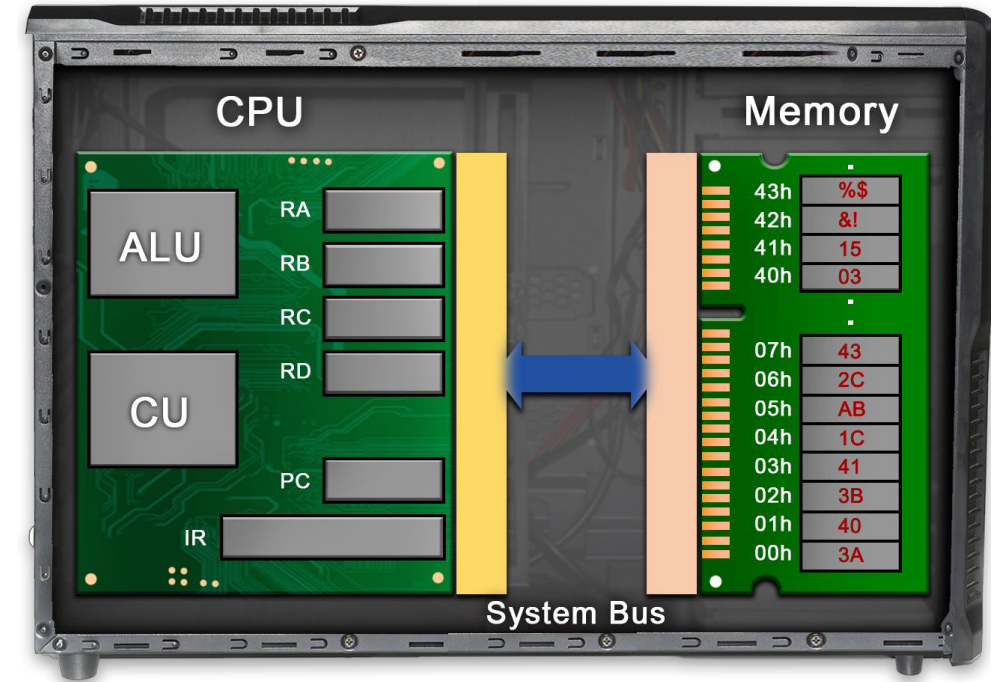
Overview of Programming Language

- Machine language
- High level programming languages
 - C, C++, Python, Java



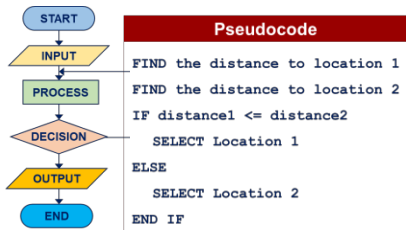
Computer Internal Operations

- Microprocessor
- Calculations performed at very high speed
 - Addition, Subtraction, Multiplication, Boolean operations



Basic Program Structure and Construct

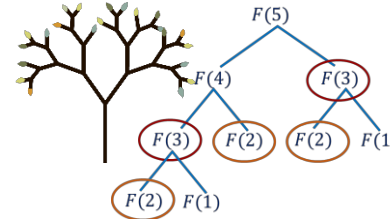
Flowchart and Pseudocode



Data Types and Variables

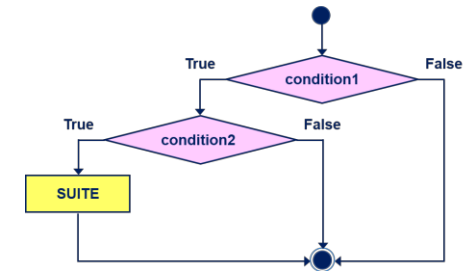


Iteration, Recursion



Selection

Branching

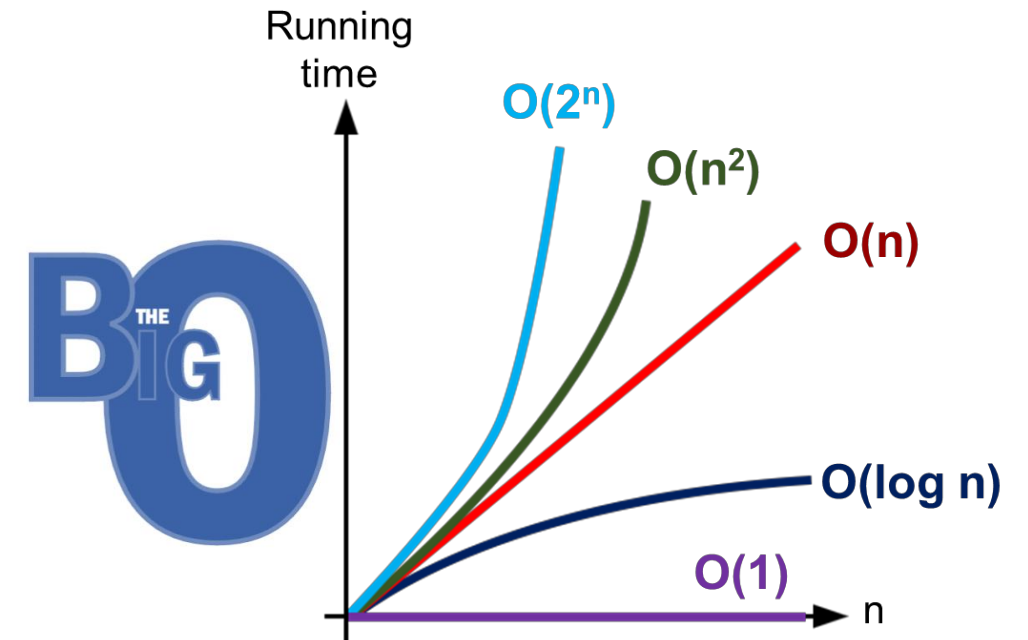


Computational Thinking Process

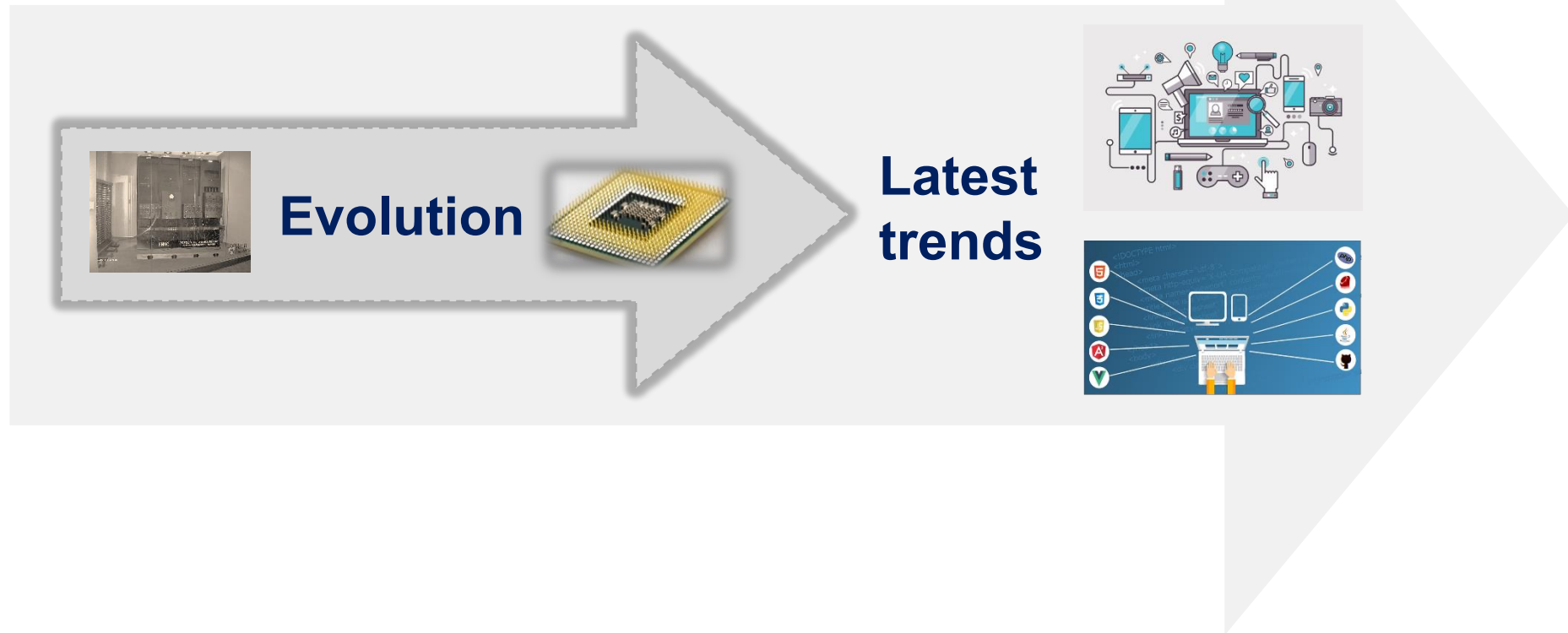


Algorithm Complexity

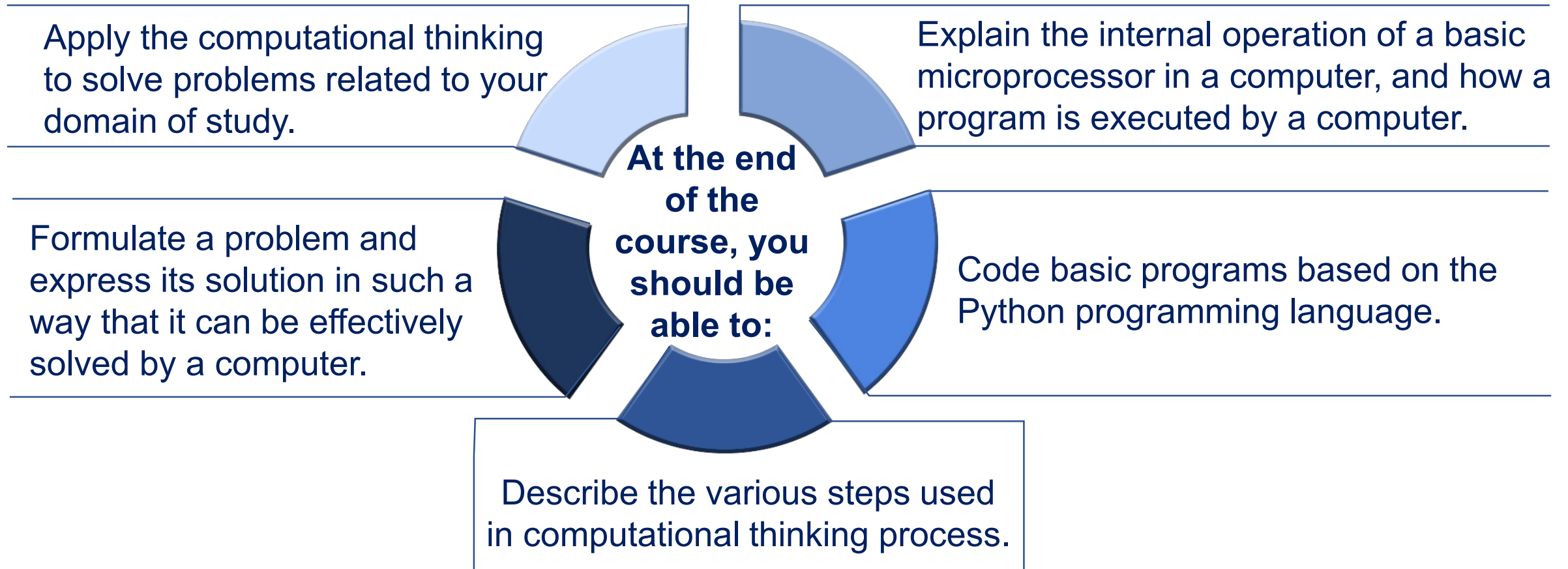
- **Big O notation**
 - to describe and compare the time complexity of algorithms
- **Scalability of algorithms**
 - input data size increases



Computing Trends



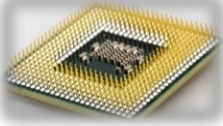
Intended Learning Outcomes



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3	4, 9		By Wolfdog1 - self-made, CC BY-SA 3.0, retrieved July 18, 2018 from https://en.wikipedia.org/w/index.php?curid=15325014 .
4	4, 10		By Jerry Mason - http://www.sitegroundwebhostingreviews.com , CC BY-SA 4.0, retrieved July 18, 2018 from https://commons.wikimedia.org/w/index.php?curid=56791665 .
5	10		By The original uploader was TexasDex at English Wikipedia. - Transferred from en.wikipedia to Commons by Andrei Stroe using CommonsHelper., CC BY-SA 3.0, retrieved July 18, 2018 from https://commons.wikimedia.org/w/index.php?curid=6557095 .

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6	10		Microprocessor [Online Image]. Retrieved July 18, 2018 from https://pixabay.com/en/amd-cpu-processor-microprocessor-1310766/ .
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